

SIRT1 Decrease Parallels the Accumulation of tau in Alzheimer Disease

Aging and metabolism-related disorders are risk factors for Alzheimer disease (AD). Since sirtuins may increase the lifespan through regulation of cellular metabolism, we compared the concentration of sirtuin 1 (SIRT1) in the brains of AD patients ($n = 19$) and controls ($n = 22$) using Western immunoblots and in situ hybridization. We report a significant reduction of SIRT1 (mRNA: -29% ; protein: -45%) in the parietal cortex of AD patients, but not in the cerebellum. Further analyses in a second cohort of 36 subjects confirmed that cortical SIRT1 was decreased in the cortex of AD patients but not in individuals with mild cognitive impairment. SIRT1 mRNA and its translated protein correlated negatively with the duration of symptoms (mRNA: $r^2 = -0.367$; protein: $r^2 = -0.326$) and the accumulation of paired helical filament tau (mRNA: $r^2 = -0.230$; protein: $r^2 = -0.119$), but weakly with insoluble amyloid- β ($A\beta_{42}$) (mRNA: $r^2 = -0.090$; protein: $r^2 = -0.072$). A significant relationship between SIRT1 levels and global cognition scores proximate to death was also found ($r^2 = +0.09$; $p = 0.049$). In contrast, cortical SIRT1 levels remained unchanged in a triple-transgenic animal model of AD. Collectively, our results indicate that loss of SIRT1 is closely associated with the accumulation of $A\beta$ and tau in the cerebral cortex of patients with AD.